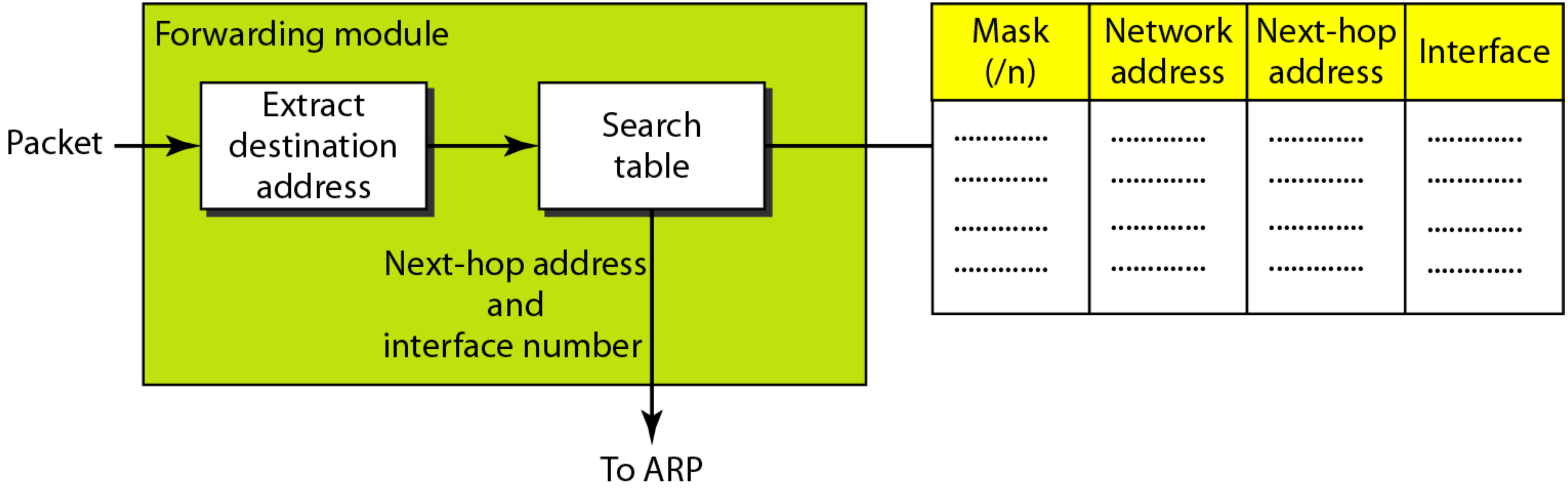
Network Layer: Routing Protocols

Dr. Alekha Kumar Mishra

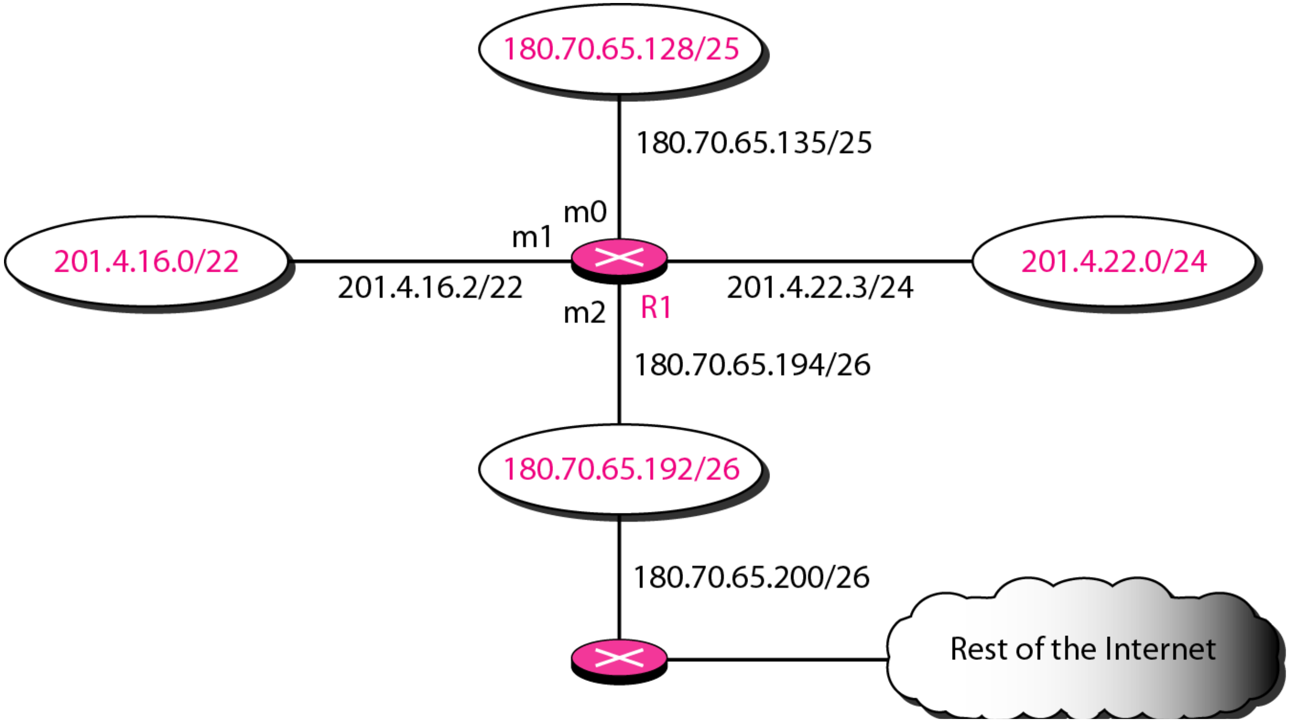
Forwarding Techniques

- Next-Hop Method Versus Route Method
  - In the next-hop method, the routing table holds only the address of the next hop instead of information about the complete route
- Network-Specific Method Versus Host-Specific Method
  - In the network-specific method, only one entry that defines the address of the destination network itself for all destination hosts
- Default Method
  - If a single next-hop router is used for almost all the destination networks (except some specific ones), instead of listing all networks in the entire Internet, a host can just have one entry called the default (normally defined as network address 0.0.0.0)

Forwarding Process

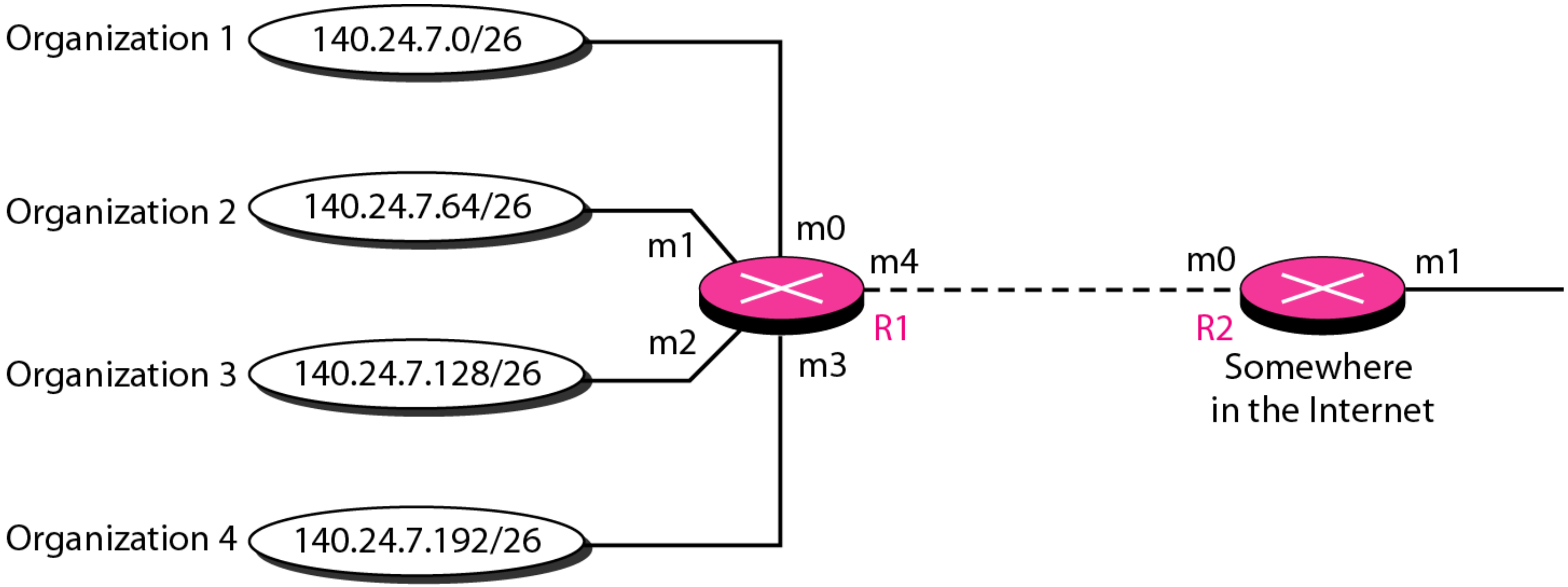


Routing table configuration (by example)



Mask	*Network Address*	*Next Hop*	*Interface*
/26	180.70.65.192	—	m2
/25	180.70.65.128	—	m0
/24	201.4.22.0	—	m3
/22	201.4.16.0	....	m1
Any	Any	180.70.65.200	m2

Address Aggregation



Mask	Network address	Next-hop address	Interface
/26	140.24.7.0	----------	m0
/26	140.24.7.64	----------	m1
/26	140.24.7.128	----------	m2
/26	140.24.7.192	----------	m3
/0	0.0.0.0	Default	m4

Routing table for R1

Mask	Network address	Next-hop address	Interface
/24	140.24.7.0	----------	m0
/0	0.0.0.0	Default	m1

Routing table for R2

Longest Mask Matching

- This principle states that the routing table is sorted from the longest mask to the shortest mask
- If there are three masks 127, 126, and 124, the mask 127 must be the first entry and 124 must be last

Routing Table

- The routing table can be either static or dynamic.
- By default all most all the tables are dynamic
- A dynamic routing table is updated periodically by using one of the dynamic routing protocols such as RIP, OSPF, or BGP
- Routing table fields
- Flags. This field defines up to five flags
  - U (up). The U flag indicates the router is up and running. If this flag is not present, it means that the router is down. The packet cannot be forwarded and is discarded.
  - G (gateway). The G flag means that the destination is in another network. The packet is delivered to the next-hop router for delivery (indirect delivery). When this flag is missing, it means the destination is in this network (direct delivery).
  - H (host-specific). The H flag indicates that the entry in the network address field is a host-specific address. When it is missing, it means that the address is only the network address of the destination.
  - D (added by redirection). The D flag indicates that routing information for this destination has been added to the host routing table by a redirection message from ICMP
  - M (modified by redirection). The M flag indicates that the routing information for this destination has been modified by a redirection message from ICMP
- Reference count - This field gives the number of users of this route at the moment.
- Use - This field shows the number of packets transmitted through this router for the corresponding destination

Mask	Network address	Next-hop address	Interface	Flags	Reference count	Use
....	....	....	....	....	....	....

Routing protocol

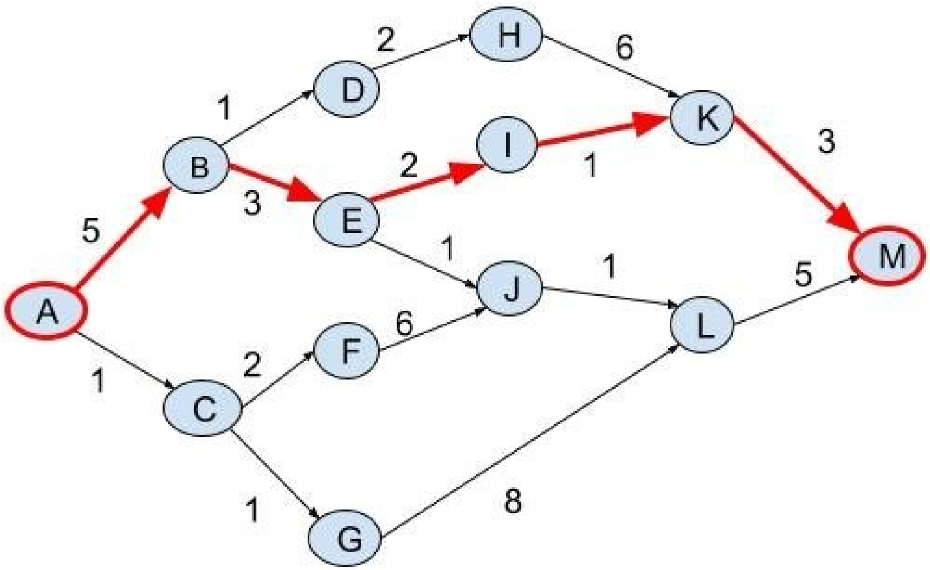
- Creates and maintain dynamic routing tables
- A routing protocol is a combination of rules and procedures that lets routers in the internet inform each other of changes
- It allows routers to share whatever they know about the internet or their neighborhood
- A router may receive information about the internet from various router, the routing protocol stores the optimal paths to a destination

Routing Algorithms

- A graph is used to formulate routing problem
- In the context of network layer routing, nodes in the graph represent routers, whereas edges in the graph connecting these nodes represent the physical link between routers
- The problem of routing a packet from source host to destination host is mapped to the problem of routing packet from source router to the destination router
- And edge in the network has a weight representing its cost
- Generally, an edge cost is measured in terms of a function of physical length, link type, link speed, and additional monetary cost associated with the link

Routing algorithms

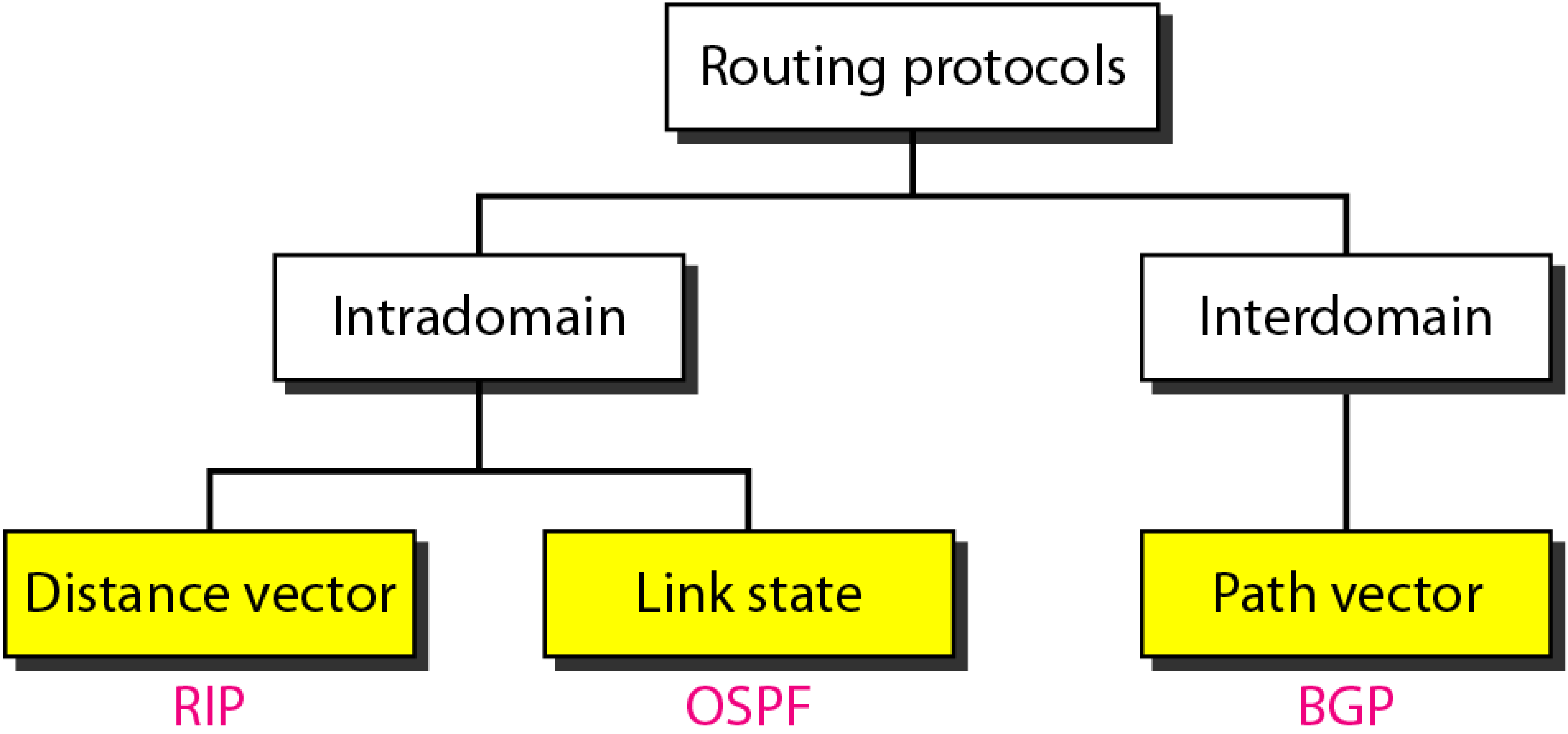
- If x and y are neighboring nodes, then (x,y) is in E(G)
- For an edge (x,y), we use the notation c(x,y) as the cost of the edge between x and y nodes
- If (x,y) is not in E(G), then c(x,y) = ∞
- The network is assumed to an undirected graph, i.e., c(x,y)=c(y,x)



Routing Algorithm

- Given a weighted graph representing a computer network, the goal of the routing algorithm(protocol) is to identify the least cost path between the source and the destination
- If a sequence of nodes (x_1, x_2, …, x_p) represent a path, then the cost of the path is simply sum of the costs of all edges along the path
  - $c(x_1,x_2)+c(x_2,x_3)+\ldots+c(x_{p-1},x_p)$
- The least cost path is also commonly known as shortest path

Routing protocols



Global routing vs Decentralized routing

- A global routing algorithm computing shortest path uses comple and global knowledge about the network. It is also called link-state routing
- A decentralized routing algorithm computes shortest path in an iterative and distributive manner. No node keeps the complete information about the network
  - Common one is Distance Vector algorithm

Distance Vector Routing

- Uses Bellman-Ford Algorithm to build and update the routing table
- Each node x begin with $D_x(y)$, and estimate of the cost of the shotest path from itselt to all the node y in the network
- Let $\mathbf{D_x}$ = [$D_x(y)$: for all y in G(V)] be the distance vector of node x
- Let $\mathbf{D_v}$ = [$D_v(y)$: for all y in G(V)] be the distance vector of each neighboring node v of x
- When a new distance vector arrives at x for a neighboring node v, then using Bellman-Ford equation, the node x update its own distance vector
- For each node y in V(G)

$$D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$$

- If distance vector has changed, node x will send its updated distance to each of its neighbor

The algorithm

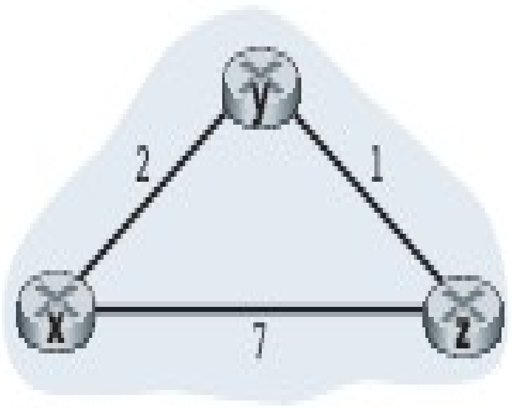
```
Distance Vector (x){
    for(each node y in V(G)){
        if( y is neighbor)
            Dx(y)=c(x,y);
        else
            Dx(y)= inf;

        for(each w in Nbr[x]){
            Dw(y)=inf;
    }
    for(each w in Nbr[x]){
        send Dx to w;
```

```
    loop:
        wait until( a change in
        linkcost to a neighbor OR a
        distance vector Dw is received
        from some w in Nbr[x]);

        for(each node y in V(G)){
          Dx(y) = min { c(x,w) + Dw(y),Dx(y) };
        if(Dx(y) has changed for any y){
            for(each w in Nbr[x])
                send Dx to w;
        }
    forever:
}
```

Example



X TABLE		
To	Cost	Next
X	0	X
Y	2	Y
Z	7	Z

Y TABLE		
To	Cost	Next
X	2	X
Y	0	Y
Z	1	Z

Z TABLE		
To	Cost	Next
X	7	X
Y	1	Y
Z	0	Z

Pass 1 - sharing

X TABLE		
To	Cost	Next
X	0	X
Y	2	Y
Z	**3**	**Y**

Y TABLE		
To	Cost	Next
X	2	X
Y	0	Y
Z	1	Z

Z TABLE		
To	Cost	Next
X	**3**	**Y**
Y	1	Y
Z	0	Z

Example 2

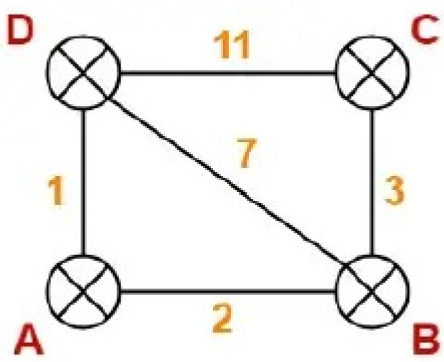


TABLE A

To	C	Nxt
A	0	A
B	2	B
C	inf	-
D	1	D

TABLE B

To	C	Nxt
A	2	A
B	0	B
C	3	C
D	7	D

TABLE C

To	C	Nxt
A	inf	-
B	3	B
C	0	C
D	11	D

TABLE D

To	C	Nxt
A	1	A
B	7	B
C	11	C
D	0	D

Pass 1

TABLE A

To	C	Nxt
A	0	A
B	2	B
C	**5**	**B**
D	1	D

TABLE B

To	C	Nxt
A	2	A
B	0	B
C	3	C
D	**3**	**A**

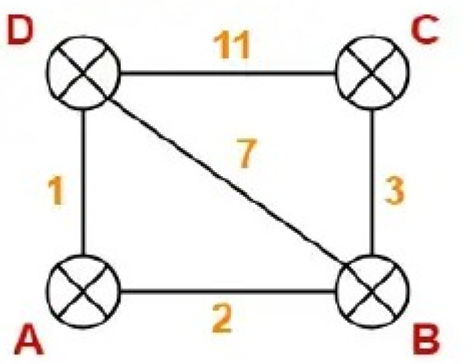
TABLE C

To	C	Nxt
A	**5**	**B**
B	3	B
C	0	C
D	**10**	**B**

TABLE D

To	C	Nxt
A	1	A
B	**3**	**A**
C	**10**	**B**
D	0	D

Example 2



Pass 1

TABLE A		
To	C	Nxt
A	0	A
B	2	B
C	**5**	**B**
D	1	D

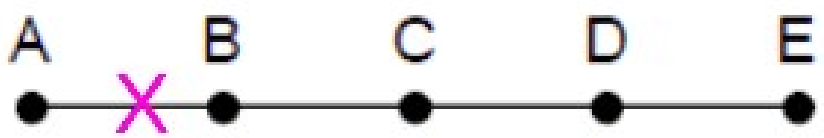
TABLE B		
To	C	Nxt
A	2	A
B	0	B
C	3	C
D	**3**	**A**

TABLE C		
To	C	Nxt
A	**5**	**B**
B	3	B
C	0	C
D	**10**	**B**

TABLE D		
To	C	Nxt
A	1	A
B	**3**	**A**
C	**10**	**B**
D	0	D

Pass 2

Count to infinity problem



B	C	D	E	
1	2	3	4	Initially
3	2	3	4	After 1 exchange
3	4	3	4	After 2 exchanges
5	4	5	4	After 3 exchanges
5	6	5	6	After 4 exchanges
7	6	7	6	After 5 exchanges
7	8	7	8	After 6 exchanges
⋮	⋮	⋮	⋮	
∞	∞	∞	∞	

RIP

- The Routing Information Protocol (RIP) is an intradomain routing protocol used inside an autonomous system
- It is based on distance vector routing
- The metric used by RIP is the number of links (networks) to reach the destination
  - The metric in RIP is called a hop count.
- Infinity is defined as 16

Link-State Routing

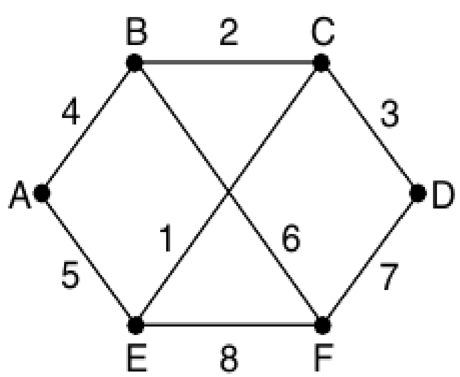
- Uses Dijkstra’s algorithm to update the table
- It computes the least-cost path from one node to all other nodes in the network
- It is iterative and has the property that after kth-pass, the least-cost or shortest path to k-destination nodes are known
- Each router must do the following (5 things)
  - Discover its neighbors and learn their network addresses
  - Set the distance or cost metric to teach of its neighbors
  - Construct a packet telling all t has just leanred
  - Sending this packet to and receive packets from all other routers
  - Compute the shortest path to every other router

Link-State Routing

- Discovering neighbors
  - Learning about neighbors is accomplished by sending a special HELLO packet on each P2P line
  - Routers on receiving the HELLO packet are expected to reply back giving its ID
- Setting the link cost
  - The cost of reaching neighbors can eb set automatically or configured by the network operator
  - Mostly, the cost is inversely propotional to bandwith
  - Delay is also added as a factor for geographically spereaded networks

Link-State Routing

- Building Link State Packets



Link State Packets

A	B	C	D	E	F
Seq.	Seq.	Seq.	Seq.	Seq.	Seq.
Age	Age	Age	Age	Age	Age
B 4	A 4	B 2	C 3	A 5	B 6
E 5	C 2	D 3	F 7	C 1	D 7
	F 6	E 1		F 8	E 8

Distribution of link-state packets

- Fundamental way is to use flooding (every incoming packet is sent to every outgoing links
- Flooding generates vast number of duplicate packets
- Approaches to minimize duplicate packets
  - Use maximum hop count
  - Keeping track of sequence number or packet ID
- Sequnce number is unique for each packet created and is incremeted for a new packet
- Issue with sequence number
  - Sequence numbers may be wrapped around
  - If routers crashes, sequence number is lost
  - Sequence number may be corrupted

Importance of age field

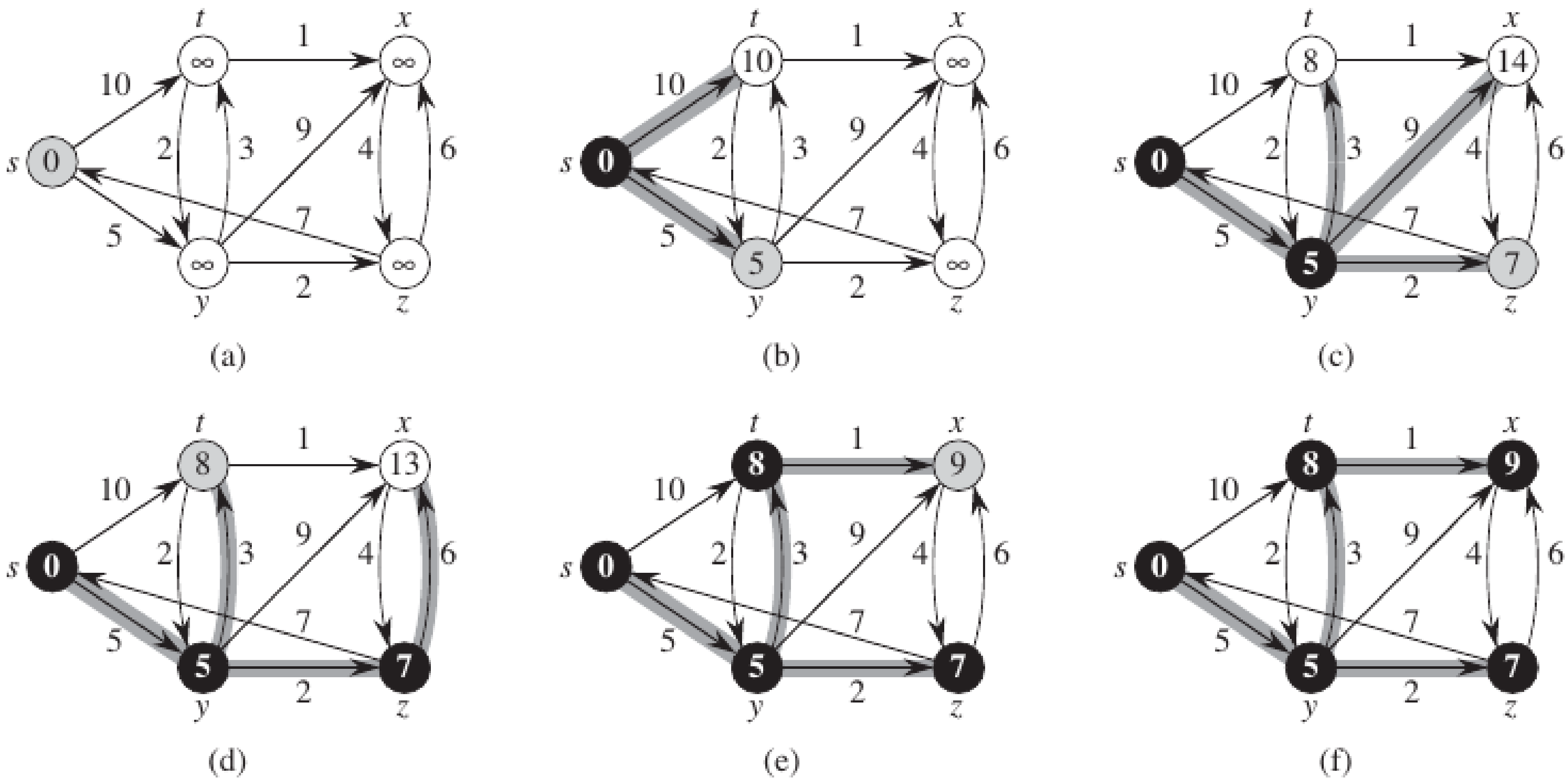
- Include age field after sequence number and decrement it once per second
- When the age its zero, the information from the router is discarded
- Age field also decremented during flooding process to make sure it does not live for indefinite period of time
- To guard against error, all link state packets are acknowledged

Link-State Routing : Computing new route

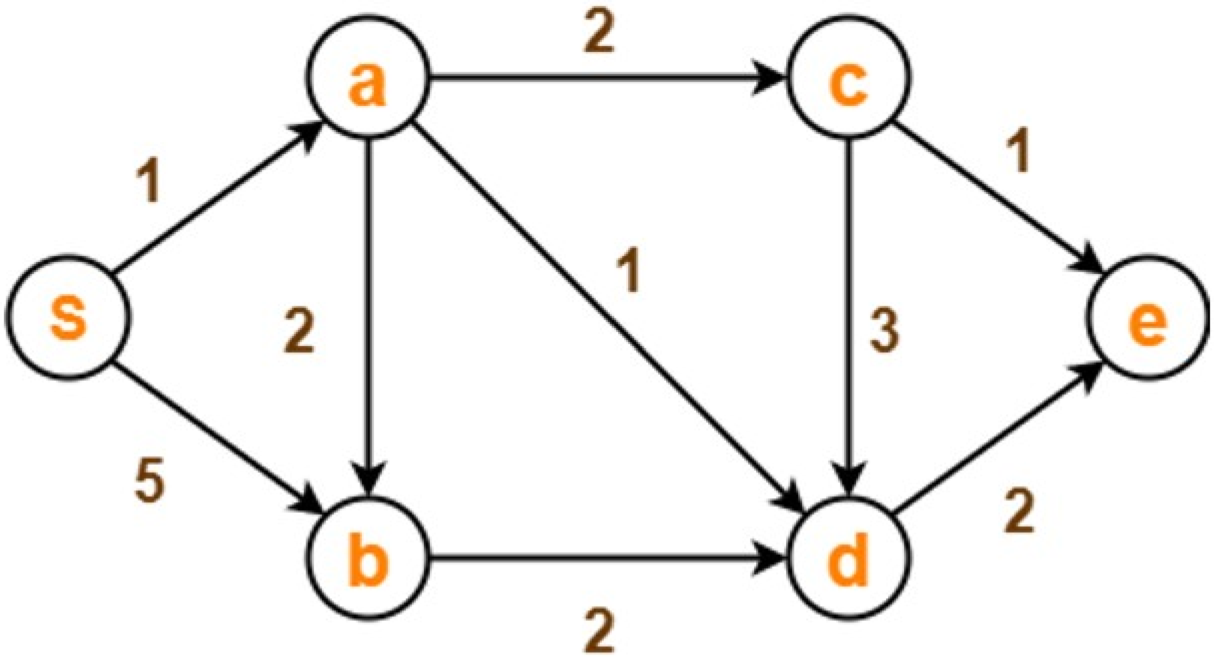
- Dijkstra's algorithm may run locally to contruct shortest path to all possible destinations

```
Dijkstra(G,c,s){
	for(each node v in V(G)){
		v.d = inf;
		v.p = NIL;
	}
	s.d=0;
	S= Φ ; //empty set
	Q=G(V) // priority queue
	while(Q is not empty){
		u=ExtractMin(Q);
		S = S U { u } ;
		for( each v in Nbr[u]){
			if(v.d > u.d + c(u,v)){
				v.d = u.d + c(u,v);
				v.p = u;
			}
		} // end of for
	}// end of while
}
```

Example 1

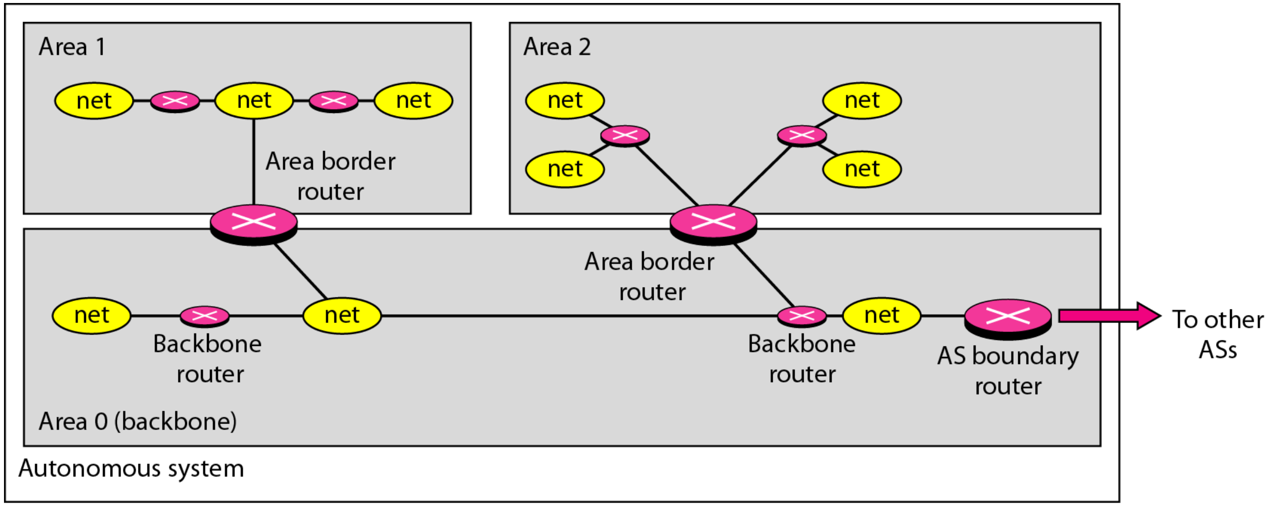


Example 2



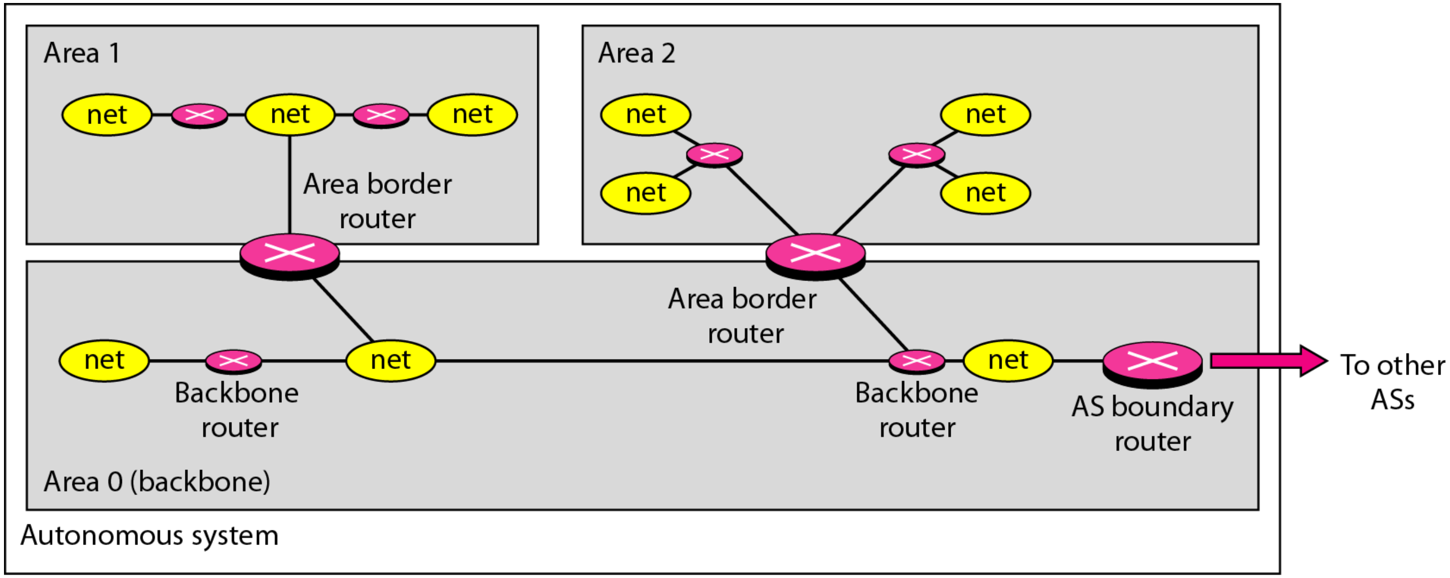
OSPF

- The Open Shortest Path First or OSPF protocol is an intradomain routing protocol
- It is also a link-state protocol that uses flooding of link-state information and Dijkstra algorithm
- An autonomous system (AS) is a group of networks and routers under the authority of a single administration
- OSPF divides an autonomous system into areas
- An area is a collection of networks, hosts, and routers all contained within an autonomous system
- All networks inside an area must be connected

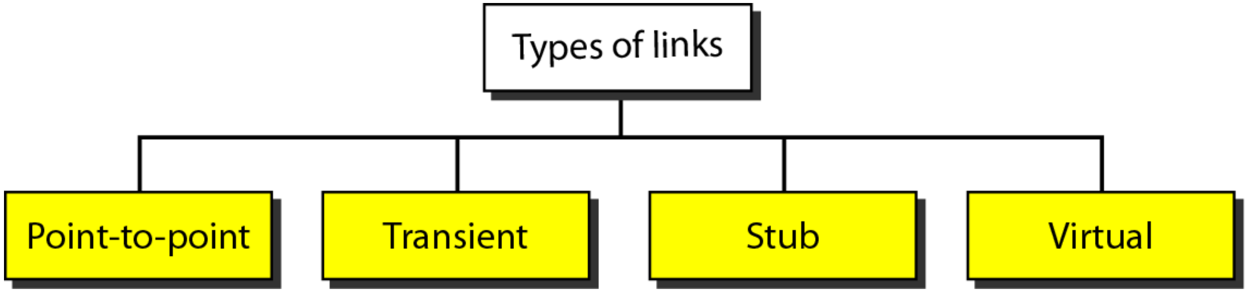


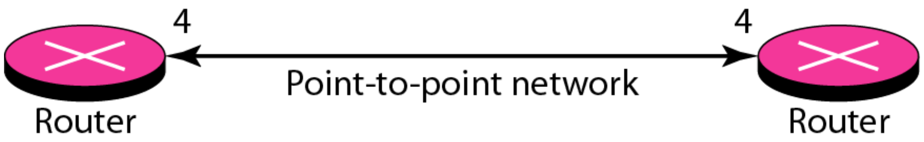
OSPF

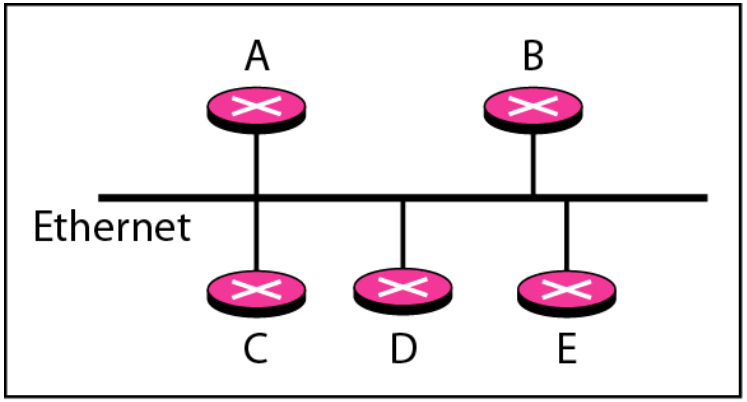
- Routers inside an area construct topological map of the area and flood the area with routing information
- At the border of an area, special routers called area border routers summarize the information about the area and send it to other areas
- A special area called the backbone; all the areas inside an autonomous system must be connected to the backbone
- The routers inside the backbone are called the backbone routers



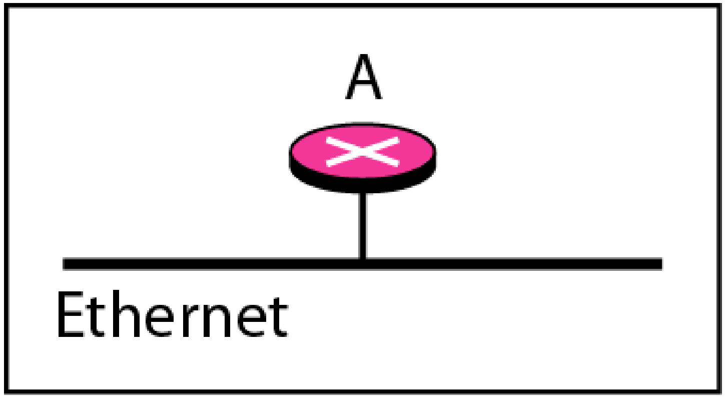
OSPF







Transient

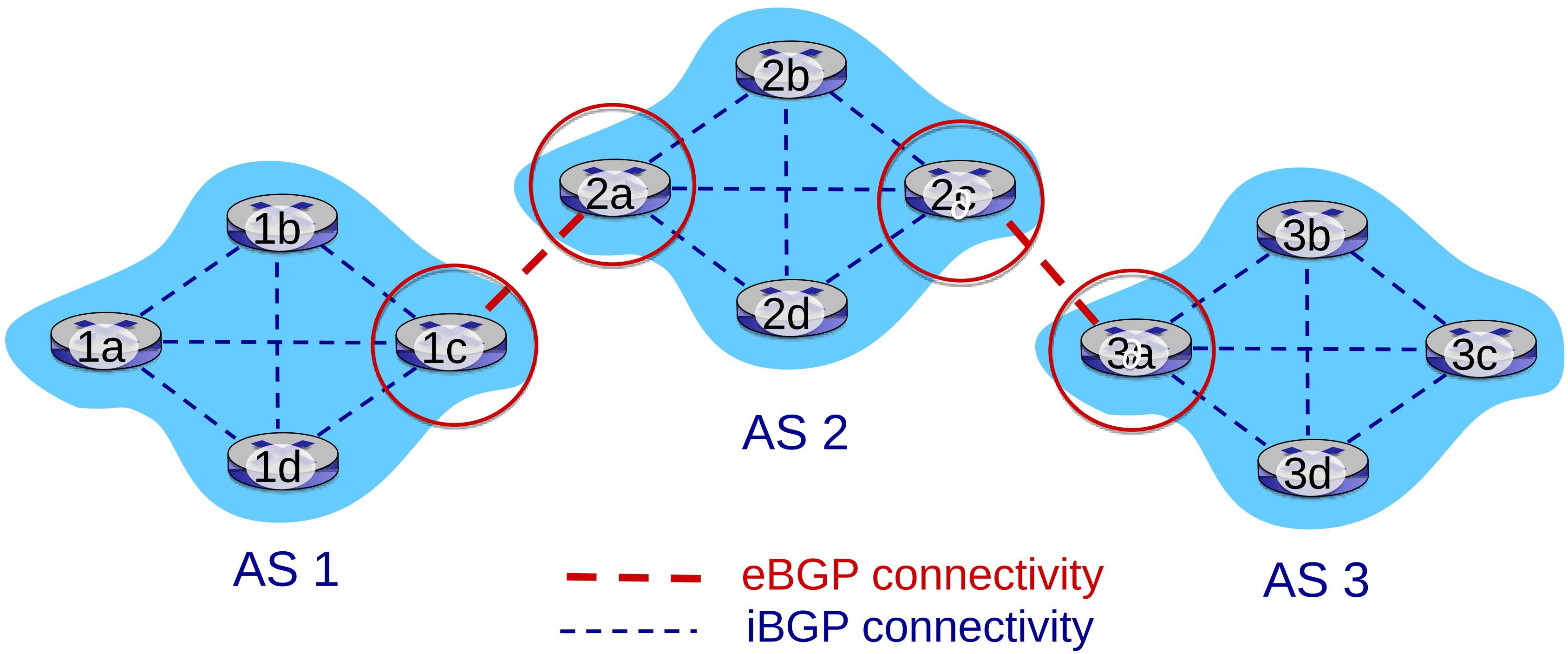


Stub

BGP

- BGP (Border Gateway Protocol) is the de facto inter-domain routing protocol
- BGP provides each AS a means to:
  - eBGP: obtain subnet reachability information from neighboring ASes
  - iBGP: propagate reachability information to all AS-internal routers.
- Determine "good" routes to other networks based on reachability information and policy
- Allows subnet to advertise its existence to rest of Internet

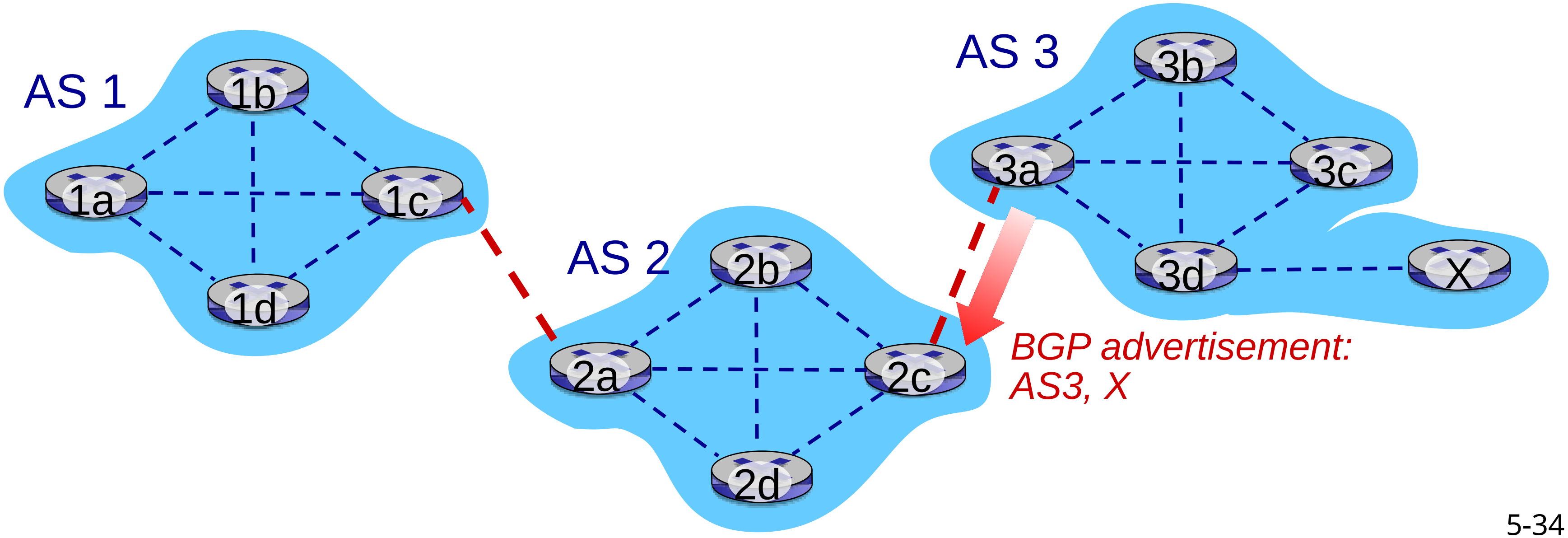
eBGP, iBGP connections



gateway routers run both eBGP and iBGP protocols

BGP Session

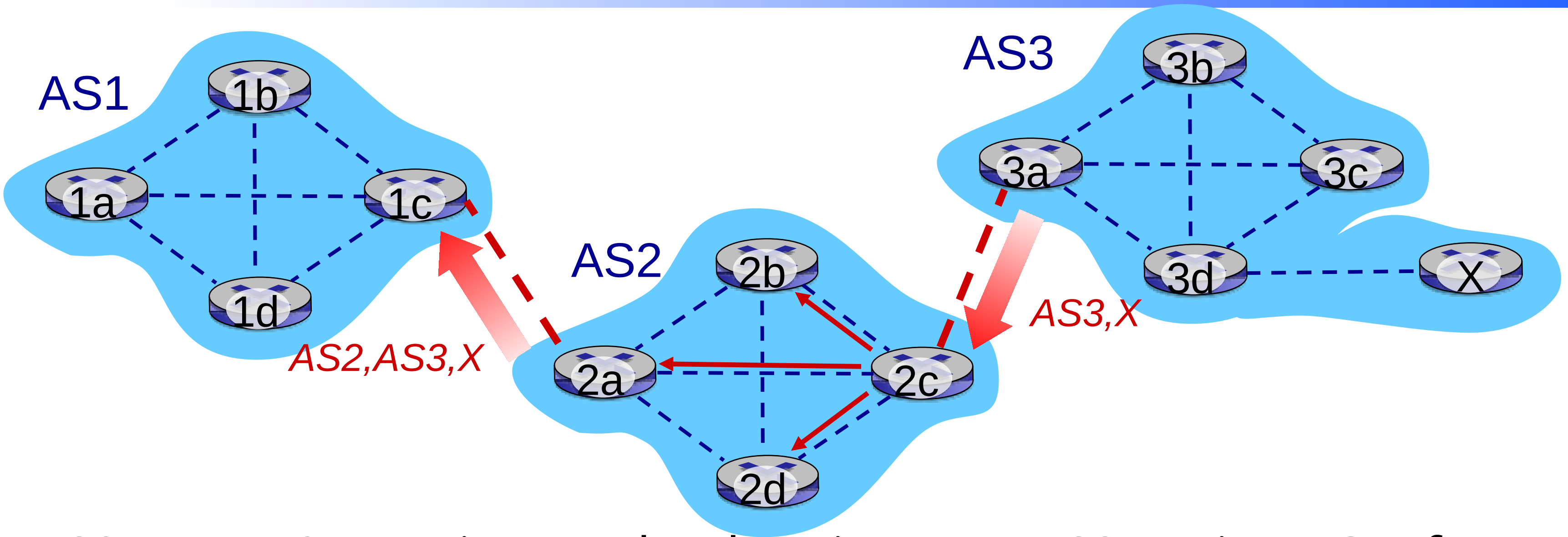
- BGP peers exchange BGP messages over BGP sessions : semi-permanent TCP connection:
  - advertising paths to different destination network prefixes
  - When AS3 gateway router 3a advertises path AS3,X to AS2 gateway router 2c, AS3 promises to AS2 it will forward datagrams towards X



BGP path attributes and routes

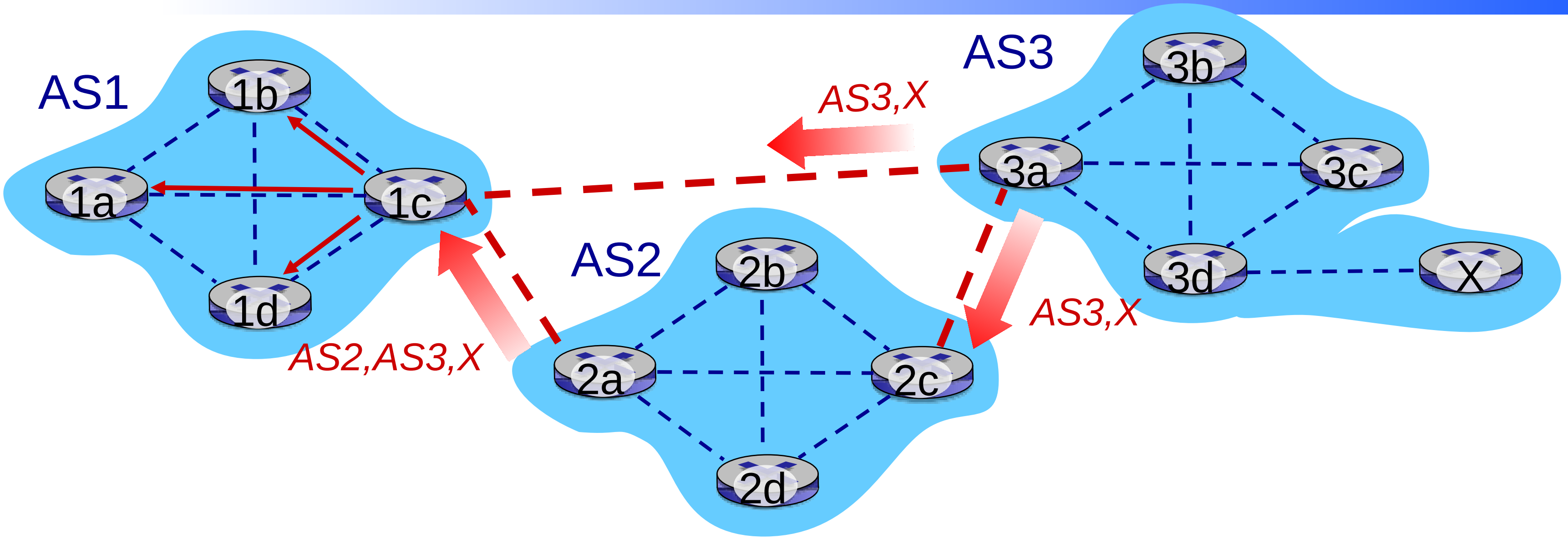
- BGP AS is identified by global Autonomous System Number (ASN)
- BGP advertised prefix includes BGP attributes
  - prefix + attributes = “route”
- Two important attributes:
  - AS-PATH: list of ASN of ASes through which prefix advertisement has passed
  - A router will reject the advertisement if AS-PATH contains its AS to avoid looping
  - NEXT-HOP: indicates specific internal-AS router to next-hop AS (begins AS-PATH)

BGP path advertisement



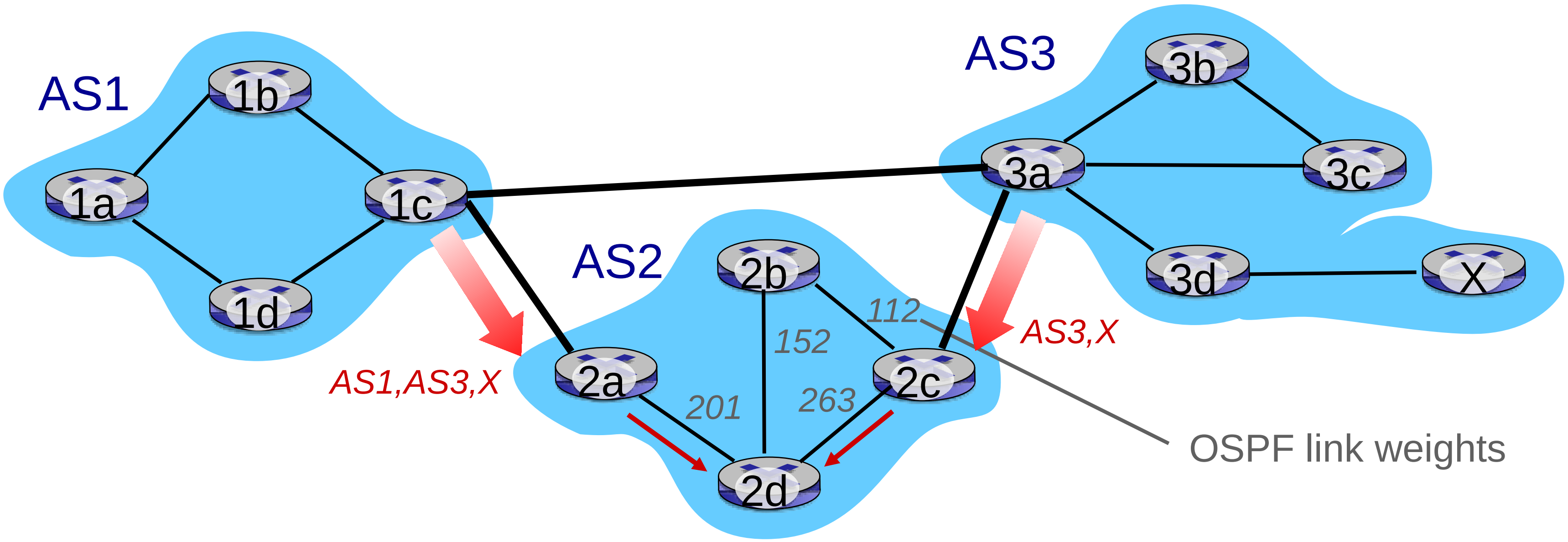
- AS2 router 2c receives path advertisement AS3,X (via eBGP) from AS3 router 3a
- Based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
- Based on AS2 policy, AS2 router 2a advertises (via eBGP) path AS2, AS3, X to AS1 router 1c

BGP path advertisement



- Gateway router may learn about multiple paths to destination:
  - AS1 gateway router 1c learns path AS2,AS3,X from 2a
  - AS1 gateway router 1c learns path AS3,X from 3a
  - Based on policy, AS1 gateway router 1c chooses path AS3,X, and advertises path within AS1 via iBGP

Hot Potato Routing



- 2d learns (via iBGP) that it can route to X via 2a or 2c
- hot potato routing: choose local gateway that has least intra-domain cost (e.g., 2d chooses 2a, even though more AS hops to X): don’t worry about inter-domain cost!

BGP Messages

- BGP messages exchanged between peers over TCP connection
- BGP messages:
  - OPEN: opens TCP connection to remote BGP peer and authenticates sending BGP peer
  - UPDATE: advertises new path (or withdraws old)
  - KEEPALIVE: keeps connection alive in absence of UPDATES; also ACKs OPEN request
  - NOTIFICATION: reports errors in previous msg; also used to close connection

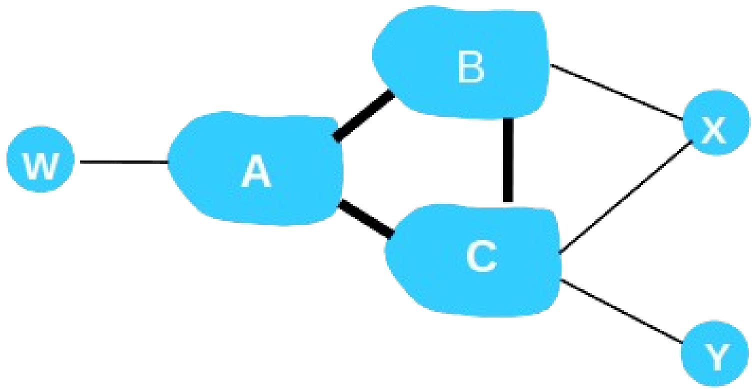
BGP Route Selection

- If there are two or more routes to the same prefix, then BGP sequentially invokes the following elimination rules until only one route remains
  - Routes are assigned local preferences value by routers, route with the highest local preference value is selected
  - In the remaining routes, the route with shortest AS-PATH is selected
  - From the remaining routes, the route with the closest NEXT-HOP router is selected
  - If more than one route remains, the router uses BGP identifier to select one

BGP Routing policy

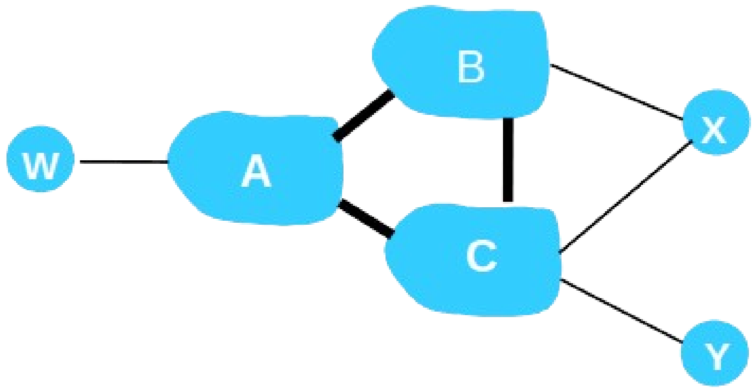
- A stub network (AS) has no knowledge of other networks
- All traffic entering AS must be destined for that network and all traffic leaving a stub network must have originated in the that network
- Selective route advertisement are done based on implemented customer/provider routing policy

BGP Routing policy



- A,B,C are provider networks
- X,W,Y are customer (of provider networks)
- X is dual-homed: attached to two networks
- Policy: X does not want to route from B to C via X .. so X will not advertise to B a route to C

BGP Routing policy



- A advertises the path Aw to B, however B chooses not to advertise the path Aw to C as C may not get required revenue from the path x C Aw

Multicast Routing

- Source-Based Tree
  - Each router needs to have one shortest path tree for each group
- Group-Shared Tree
  - Only one designated router, called the center core, takes the responsibility of distributing multicast traffic
  - The core has m shortest path trees in its routing table
  - If a non-core router receives a multicast packet, it encapsulates the packet in a unicast packet and sends it to the core router

Multicast Link State Routing: MOSPF

- MOSPF protocol is an extension of the OSPF protocol that uses multicast link state routing to create source-based trees
- When a router receives link states, it creates n (n is the number of groups) topologies, from which n shortest path trees are made by using Dijkstra's algorithm
- Each router has a routing table that represents as many shortest path trees as there are groups.
- Requires a new link state update packet to associate the unicast address of a host with the group address or addresses the host is sponsoring
- For efficiency, the router calculates the shortest path trees on demand

Self Study

- Multicast Distance Vector: DVMRP

Next : Transport Layer